

Start here X ds 12.c X

```
1  #include <stdio.h>
2
3  #define MAX 10
4
5  int graph[MAX][MAX];
6  int visited[MAX];
7  int queue[MAX];
8  int front = -1, rear = -1;
9  int vertices;
10
11 // Queue operations for BFS
12 void enqueue(int value)
13 {
14     queue[++rear] = value;
15 }
16
17 int dequeue()
18 {
19     return queue[++front];
20 }
21
22 // BFS Traversal
23 void BFS(int start)
24 {
25     int i;
```

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```
25     int i;
26
27     printf("BFS Traversal: ");
28
29     visited[start] = 1;
30     enqueue(start);
31
32     while(front != rear)
33     {
34         int node = dequeue();
35
36         printf("%d ", node);
37
38         for(i = 0; i < vertices; i++)
39         {
40             if(graph[node][i] == 1 && visited[i] == 0)
41             {
42                 visited[i] = 1;
43                 enqueue(i);
44             }
45         }
46     }
47 }
48
49 // DFS Traversal
50 void DFS(int start)
```

```

49 // DFS Traversal
50 void DFS(int start)
51 {
52     int i;
53
54     printf("%d ", start);
55
56     visited[start] = 1;
57
58     for(i = 0; i < vertices; i++)
59     {
60         if(graph[start][i] == 1 && visited[i] == 0)
61         {
62             DFS(i);
63         }
64     }
65 }
66
67 int main()
68 {
69     int edges;
70     int i, src, dest;
71     int start;
72
73     printf("Enter number of vertices: ");
74     scanf("%d", &vertices);
75

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```
76 // Initialize graph
77 for(i = 0; i < vertices; i++)
78 {
79     visited[i] = 0;
80
81     for(int j = 0; j < vertices; j++)
82     {
83         graph[i][j] = 0;
84     }
85 }
86
87 printf("Enter number of edges: ");
88 scanf("%d", &edges);
89
90 // Input edges
91 for(i = 0; i < edges; i++)
92 {
93     printf("Enter edge (source destination): ");
94     scanf("%d %d", &src, &dest);
95
96     graph[src][dest] = 1;
97     graph[dest][src] = 1; // Undirected graph
98 }
99
100 printf("Enter starting vertex: ");
101 scanf("%d", &start);
102
```

```
94         scanf("%d %d", &src, &dest);
95
96         graph[src][dest] = 1;
97         graph[dest][src] = 1; // Undirected graph
98     }
99
100     printf("Enter starting vertex: ");
101     scanf("%d", &start);
102
103     // BFS
104     BFS(start);
105
106     // Reset visited array for DFS
107     for(i = 0; i < vertices; i++)
108     {
109         visited[i] = 0;
110     }
111
112     printf("\nDFS Traversal: ");
113
114     // DFS
115     DFS(start);
116
117     return 0;
118 }
119
```

```
"C:\Users\darsh\OneDrive\  ×  +  ▾  
Enter number of vertices: 5  
Enter number of edges: 4  
Enter edge (source destination): 0 1  
Enter edge (source destination): Enter edge (source destination): Enter starting vertex  
: BFS Traversal: 0 1  
DFS Traversal: 0 1  
Process returned 0 (0x0)   execution time : 23.246 s  
Press any key to continue.
```